

Questions with Answer Keys

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Q1 - Chord, Focal Chord and Its Properties.

If Y -axis is the directrix of the ellipse with eccentricity $e = 1/2$ and corresponding focus is at $(3, 0)$, then the equation to its auxiliary circle is

(1) $x^2 + y^2 - 8x + 12 = 0$

(2) $x^2 + y^2 - 8x - 12 = 0$

(3) $x^2 + y^2 - 8x + 9 = 0$

(4) $x^2 + y^2 = 4$

Q2 - Chord, Focal Chord and Its Properties.

The locus of the middle points of chords of an ellipse $\frac{x^2}{16} + \frac{y^2}{25} = 1$, passing through $P(0, 5)$ is another ellipse E . The coordinates of foci of the ellipse E is

(1) $\left(0, \frac{3}{5}\right)$ and $\left(0, -\frac{3}{5}\right)$

(2) $(0, -4)$ and $(0, 1)$

(3) $(0, 4)$ and $(0, 1)$

(4) $\left(0, \frac{11}{2}\right)$ and $\left(0, -\frac{1}{2}\right)$

Q3 - Chord, Focal Chord and Its Properties.

The eccentric angle of the extremities of latusrecta of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is

(1) $\tan^{-1}\left(\pm \frac{b}{ae}\right)$

(2) $\tan^{-1}\left(\pm \frac{a}{be}\right)$

(3) $2 \tan^{-1}\left(\pm \frac{b}{ae}\right)$

(4) None of these

Q4 - Chord, Focal Chord and Its Properties.

Let α and β be the eccentric angles of the extremities of a focal chord of an ellipse of eccentricity e , then

$$\tan \frac{\alpha}{2} \cdot \tan \frac{\beta}{2} \text{ is}$$

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(1) $-\left(\frac{1 \mp e}{1 \mp e}\right)$

(2) $\left(\frac{1+\theta}{1-\theta}\right)$

(3) $-\left(\frac{1 \mp e}{1 \pm e}\right)$

(4) None of these

Q5 - Chord, Focal Chord and Its Properties.

If any two chords can be drawn through two points on the major axis of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ equidistant from the centre, then the value of $\tan \frac{\alpha}{2} \cdot \tan \frac{\beta}{2} \cdot \tan \frac{\gamma}{2} \tan \frac{\delta}{2}$ is (where $\alpha, \beta, \gamma, \delta$ are the eccentric angles of extremities of the chords)

(1) 0

(2) 1

(3) -1

(4) $1/2$

Q6 - Tangent and its properties

The equations of tangents to the ellipse $3x^2 + 4y^2 = 12$, which are perpendicular to $y + 2x = 4$ is

(1) $2y = x \pm 3$

(2) $2y = x \pm 1$

(3) $2y = x \pm 4$

(4) None of these

Q7 - Tangent and its properties

The straight line $x + 2y - 4 = 0$ touches the ellipse $3x^2 + 4y^2 = 12$ at

(1) (1,3)

(2) $(1/2, 3/2)$

(3) $(0, \sqrt{3})$

(4) $(1, 3/2)$

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Q8 - Tangent and its properties

The locus of the mid-points of the portion of tangents to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ intercepted between the axes is

(1) $a^2x^2 + b^2y^2 = x^2y^2$

(2) $a^2x^2 + b^2y^2 = 4x^2y^2$

(3) $a^2y^2 + b^2x^2 = 4x^2y^2$

(4) $a^2y^2 + b^2x^2 = x^2y^2$

Q9 - Tangent and its properties

The locus of foot of the perpendicular on a varying tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, from either of its foci is

(1) $x^2 + y^2 = a^2 + b^2$

(2) $x^2 + y^2 = a^2$

(3) $x^2 + y^2 = b^2$

(4) $x^2 + y^2 = 2a^2$

Q10 - Tangent and its properties

The radius of the largest circle with centre (1,0) that can be inscribed in the ellipse $x^2 + 4y^2 = 16$ is

(1) $\frac{11}{3}$

(2) $\frac{11}{\sqrt{3}}$

(3) $\frac{\sqrt{11}}{3}$

(4) $\sqrt{\frac{11}{3}}$

Q11 - Tangent and its properties

If $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b$) and $x^2 - y^2 = c^2$ cut at right angles, then

(1) $a^2 + b^2 = 2c^2$

(2) $b^2 - a^2 = 2c^2$

(3) $a^2 - b^2 = 2c^2$

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(4) $a^2b^2 = 2c^2$

Q12 - Tangent and its properties

A common tangent to $9x^2 - 16y^2 = 144$ and $x^2 + y^2 = 9$ is

(1) $y = \frac{3}{\sqrt{7}}x + \frac{15}{\sqrt{7}}$

(2) $y = 3\sqrt{\frac{2}{7}}x + \frac{15}{\sqrt{7}}$

(3) $y = \sqrt{\frac{3}{7}}x + 15\sqrt{7}$

(4) None of these

Q13 - Tangent and its properties

The locus of point of intersection of two tangents of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, if the product of their slopes is c^2 , will be

(1) $y^2 - b^2 = c^2(x^2 + a^2)$

(2) $y^2 + b^2 = c^2(x^2 - a^2)$

(3) $y^2 + a^2 = c^2(x^2 - b^2)$

(4) $y^2 - a^2 = c^2(x^2 + b^2)$

Q14 - Tangent and its properties

The locus of point of intersection of two tangents to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, the sum of whose slopes is c^2 , is

(1) $(x^2 - a^2)c^2 = y^2 - b^2$

(2) $(x^2 + a^2)c^2 = y^2 + b^2$

(3) $(x^2 - a^2)c^2 = 2xy$

(4) $(x^2 + a^2)c^2 = y^2 - b^2$

Q15 - Normal and its properties

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PQ is a double ordinate of the ellipse $x^2 + 9y^2 = 9$, the normal at P meets the diameter through Q at R , then the locus of mid-point of PR is

- (1) a circle
- (2) a parabola
- (3) an ellipse
- (4) a hyperbola

Q16 - Normal and its properties

The area of rectangle formed by the perpendiculars from the centre of the standard ellipse to the tangent and normal at its point whose eccentric angle is $(\pi/4)$, is

- (1) $\frac{(a^2-b^2)ab}{a^2+b^2}$
- (2) $\frac{(a^2+b^2)ab}{a^2-b^2}$
- (3) $\frac{a^2-b^2}{ab(a^2+b^2)}$
- (4) $\frac{a^2+b^2}{(a^2-b^2)ab}$

Q17 - Normal and its properties

If the normal at an end of a latusrectum of an ellipse passes through one extremity of the minor axis, then the eccentricity of ellipse is given by

- (1) $e^4 - e^2 + 1 = 0$
- (2) $e^4 + e^2 - 1 = 0$
- (3) $e^2 + e - 1 = 0$
- (4) None of these

Q18 - Normal and its properties

Any ordinate MP of an ellipse meets the auxiliary circle in θ . Then, the locus of the point of intersection of the normals at P and Q is

- (1) $x^2 + y^2 = a^2 + b^2$

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(2) $x^2 + y^2 = (a + b)^2$

(3) $x^2 + y^2 = a^2 - b^2$

(4) $x^2 + y^2 = (a - b)^2$

Q19 - Normal and its properties

Let $P(a \sec \theta, b \tan \theta)$ and $Q(a \sec \phi, b \tan \phi)$, where $\theta + \phi = \frac{\pi}{2}$, be two points on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If (h, k) is the point of intersection of the normals at P and Q , then k is equal to

(1) $\frac{a^2+b^2}{a}$

(2) $\frac{a^2+b^2}{b}$

(3) $-\left(\frac{a^2+b^2}{a}\right)$

(4) $-\left(\frac{a^2+b^2}{b}\right)$

Q20 - Normal and its properties

The number of normals which can be drawn from an external point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, is

(1) 1

(2) 4

(3) 6

(4) 8

Q21 - Normal and its properties

If the tangent and normal to $x^2 - y^2 = 4$ at a point cut off intercepts a_1, a_2 on the X -axis respectively and b_1, b_2 on Y -axis respectively, then the value of $a_1 a_2 + b_1 b_2$ is

(1) 1

(2) -1

(3) 0

(4) 4

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Q22 - Normal and its properties

The locus of feet of the normals drawn from a point to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ lie on the curve

(1) $a^2y(x+h) + b^2x(y+k) = 0$

(2) $b^2y(x+h) + a^2x(y+k) = 0$

(3) $a^2y(x-h) + b^2x(y-k) = 0$

(4) $b^2y(x-h) + a^2x(y-k) = 0$

Q23 - Chord of Contact, Diameter and Conjugate Diameter

The chord of contact of tangents drawn from the point (h, k) to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ will subtend a right angle at the centre, if $\frac{h^2}{a^4} + \frac{k^2}{b^2}$ is

(1) $a^2 + b^2$

(2) $\frac{1}{a^2} + \frac{1}{b^2}$

(3) $\frac{a^2+b^2}{ab}$

(4) None of these

Q24 - Chord of Contact, Diameter and Conjugate Diameter

The middle point of chord intercepted on the line $2x - y + 3 = 0$ by the ellipse $\frac{x^2}{10} + \frac{y^2}{6} = 1$ is

(1) $\left(\frac{30}{23}, -\frac{9}{23}\right)$

(2) $\left(-\frac{30}{23}, \frac{9}{23}\right)$

(3) $\left(-\frac{30}{23}, -\frac{9}{23}\right)$

(4) $\left(\frac{30}{23}, \frac{9}{23}\right)$

Q25 - Chord of Contact, Diameter and Conjugate Diameter

The locus of the point of intersection of tangents at the end-points of conjugate diameter of the ellipse

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, is

(1) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 4$

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(2) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 3$

(3) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 2$

(4) None of these

Q26 - Chord of Contact, Diameter and Conjugate Diameter

CP and CD are conjugate diameters of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Then, the locus of mid-point of PD is

(1) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{1}{2}$

(2) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

(3) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{3}{2}$

(4) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 2$

Q27 - Chord of Contact, Diameter and Conjugate Diameter

CP and CD are conjugate semi-diameters of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the circles with CP and CD as diameters intersect at R . Then, R lies on

(1) $2(x^2 + y^2) = a^2x^2 + b^2y^2$

(2) $2(x^2 + y^2)^2 = a^2 + b^2$

(3) $2(x^2 + y^2)^2 = a^2x^2 + b^2y^2$

(4) None of the above

Q28 - Asymptotes and Rectangular Hyperbola

The angle between the asymptotes of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is

(1) $2 \tan^{-1}\left(\frac{b}{a}\right)$

(2) $\tan^{-1}\left(\frac{a}{b}\right)$

(3) $2 \tan^{-1}\left(\frac{a}{b}\right)$

(4) $\tan^{-1}\left(\frac{b}{a}\right)$

Q29 - Asymptotes and Rectangular Hyperbola

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The product of length of perpendiculars drawn from any point on the hyperbola $x^2 - 2y^2 - 2 = 0$ to its asymptotes, is

- (1) $1/2$
- (2) $2/3$
- (3) $3/2$
- (4) 2

Q30 - Asymptotes and Rectangular Hyperbola

The equation of hyperbola whose asymptotes are the straight lines $3x - 4y + 7 = 0$ and $4x + 3y + 1 = 0$ and which passes through the origin, is

- (1) $(3x - 4y + 7)(4x + 3y + 1) + 7 = 0$
- (2) $(3x - 4y + 7)(4x + 3y + 1) - 7 = 0$
- (3) $(3x - 4y + 7)(4x + 3y - 1) + 7 = 0$
- (4) $(3x + 4y + 7)(4x - 3y + 1) = 0$

Q31 - Asymptotes and Rectangular Hyperbola

The asymptotes of a hyperbola having centre at the point $(1,2)$ are parallel to the lines $2x + 3y = 0$ and $3x + 2y = 0$. If the hyperbola passes through the point $(5,3)$, its equation is

$(2x + 3y - 8)(3x + 2y + 7) = \lambda$, then λ is equal to

- (1) 148
- (2) 152
- (3) 154
- (4) 160

Q32 - Asymptotes and Rectangular Hyperbola

A rectangular hyperbola whose centre is C , is cut by any circle of radius r in four points P, Q, R and S . Then, $CP^2 + CQ^2 + CR^2 + CS^2$ is

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(1) r^2

(2) $2r^2$

(3) $3r^2$

(4) $4r^2$

Q33 - Asymptotes and Rectangular Hyperbola

If e is the eccentricity of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and θ be the angle between the asymptotes, then $\sec(\theta/2)$ is equal to

(1) $2e$

(2) e^2

(3) e

(4) None of these

Q34 - Asymptotes and Rectangular Hyperbola

If normal to the rectangular hyperbola $xy = c^2$ at the point t on it intersects the hyperbola at t_1 , then $t^3 t_1$ is equal to

(1) 1

(2) 2

(3) -1

(4) -2

Q35 - Asymptotes and Rectangular Hyperbola

If the normal at the point $P(x_i, y_i)$, $i = 1, 2, 3, 4$ on the hyperbola $xy = c^2$ are concurrent at the point $Q(h, k)$, then $\frac{(x_1+x_2+x_3+x_4)(y_1+y_2+y_3+y_4)}{x_1x_2x_3x_4}$ is

(1) $\frac{hk}{c^2}$

(2) $\frac{hk}{c^4}$

(3) $-\frac{hk}{c^2}$

(4) $-\frac{hk}{c^4}$

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Answer Key

Q1 (1)	Q2 (3)	Q3 (1)	Q4 (3)
Q5 (2)	Q6 (3)	Q7 (4)	Q8 (3)
Q9 (2)	Q10 (4)	Q11 (3)	Q12 (2)
Q13 (2)	Q14 (3)	Q15 (3)	Q16 (1)
Q17 (2)	Q18 (2)	Q19 (4)	Q20 (2)
Q21 (3)	Q22 (3)	Q23 (2)	Q24 (2)
Q25 (3)	Q26 (1)	Q27 (3)	Q28 (1)
Q29 (2)	Q30 (2)	Q31 (3)	Q32 (4)
Q33 (3)	Q34 (3)	Q35 (4)	